

Institute of Actuaries of India

Subject SA2 – Life Insurance

June 2019 Examination

INDICATIVE SOLUTION

Introduction

The indicative solution has been written by the Examiners with the aim of helping candidates. The solutions given are only indicative. It is realized that there could be other points as valid answers and examiner have given credit for any alternative approach or interpretation which they consider to be reasonable.

Solution 1:

[2 marks for each point appropriately discussed]

i) The possible circumstances under which gross premium reserves could be higher than the underlying asset shares are set out below:

Timing

At early policy durations, asset shares are typically low or may even be negative. This is generally attributable to the high initial expenses. For recently launched products wherein most policies will be at early durations, the gross premium reserves may be higher than the asset shares.

Guarantees

It is possible that certain products carry onerous guarantees relative to the prevailing interest rates. For instance, old generation products may have been priced at significantly higher interest rates. For such products, the underlying asset shares may not be sufficient to cover the guaranteed benefits, including vested reversionary bonuses. Under such circumstances, the gross premium reserves would be higher than the asset shares.

Investment performance

If the investment strategy entails significant proportion of risky assets e.g. equities or property and the recent investment performance of such risky assets has been poor, this could lead to a fall in asset shares. In comparison, the progression of the gross premium reserves tends to be more stable as it takes time to adjust the future bonuses, having regard to the policyholders' reasonable expectations ("PRE"). Under such circumstances, the gross premium reserves may be higher than the asset shares.

Also, it is important to ensure that the asset shares are re-computed on a 'book value' basis (i.e. ignoring the unrealized gains / losses), before comparing against the gross premium reserves. In the absence of this, any unrealized losses reflected in asset shares may result in the gross premium reserves being higher than the asset shares.

Over-distribution of surplus

If there have been cross-subsidies or inter-generational inequity in the historical bonus declarations, then this would lead to a situation wherein certain cohorts received higher bonuses than supportable by their respective asset shares. This could lead to the gross premium reserves being for such cohorts / products being higher than the underlying asset shares.

Valuation basis

The gross premium valuation requires reflecting future bonuses to be consistent with the valuation interest rate. If the allowance for future bonuses reflected in the valuation is overly prudent, then this would lead to the gross premium reserves being higher than the asset shares.

Likewise, if the valuation basis is generally more prudent (e.g. not taking into account lapses in the statutory valuation even if the bonus management framework does not recycle lapse / surrender profits into the asset shares of in-force policies; or adopting large margins for adverse deviations in the expense or mortality bases etc.), then this would lead to the gross premium reserves being higher than the asset shares.

Reduced paid-up policies

After the premiums are discontinued on policies that have acquired surrender values, such policies continue to be in-force in a reduced paid-up form. Such policies no longer participate in the future performance of the with-profits fund (although the bonuses already vested are guaranteed and reflected in the reduced paid-up benefits). Whilst a prospective reserve is set up towards such reduced paid-up policies, their asset shares may no longer be tracked, especially if these are not re-cycled into the asset shares of the remaining in-force

policies. If such policies are not excluded from the comparison of asset share vs. gross premium reserves, they could skew the results and show prospective reserves as being higher than the asset shares. [9]

ii) Various factors that the Company will need to consider for implementing the in-force illustrations as a tool to manage the PREs are set out below: [0.5 marks for any suitable point made and appropriately discussed]

New business illustrations are relatively straightforward as they are produced once at the point of sale and based on illustrated bonus rates that are consistent with the pricing assumptions and the assumed gross investment returns of 4% p.a. and 8% p.a. In comparison, in-force illustrations are more complex as they would need to reflect historical transactions (e.g. historically declared bonus rates, any policyholder transactions such as changes in sum assured, change in premiums etc.) in addition to allowing for the future projected bonuses. Typically, producing in-force illustrations would require significant investments in the IT systems. If the Company has been selling 'pensions' business in the past (under the IRDAI products regulations issued in 2013), it should have already implemented the in-force benefit illustrations in its IT systems. If not, the Company has to consider whether it has the resources and financial capacity to make such an investment. [2]

In-force illustrations would significantly influence the PREs. The Company would need to ensure it has established appropriate processes and controls around them. It would be desirable to have the same rigor around in-force illustrations as exists for new business illustrations (e.g. Appointed Actuary's sign-off and compliance with Actuarial Practice Standard 5 (APS5), Board approved illustration basis etc.) [2]

The Company should study the general market practice in relation to in-force illustrations. If most other insurers in the market provide such illustrations, then the Company's policyholders would also reasonably expect to receive those. [0.5]

The Company should consider the frequency of producing such illustrations. Producing them annually would seem a natural choice and indeed consistent with the requirements specified in the IRDA's product regulations for 'pensions' business. However, if the bonus rates tend to be stable from one year to another, then producing the illustrations every two to three years may also be acceptable (for 'non-pension' policies). [1]
[5]

iii) Key requirements of the APS-5 in relation to the benefit illustrations for with-profits products are set out below: [0.5 mark for each relevant point included in APS-5]

- Illustrations must be prepared in consultation with the Appointed Actuary and authorized for use by the management of a life insurer.
- Guaranteed benefits
 - These should be clearly distinguished from the non-guaranteed benefits. The circumstances under which the guaranteed benefits would become payable and appropriate conditions, if any, should be clearly stated.
- Non-guaranteed benefits
 - The illustration for non-guaranteed benefits shall be based on assumptions about future experience such as investment return, taxation, mortality, morbidity, charges, expenses, terminations by lapses or surrenders or discontinuance of premiums leading to a policy becoming reduced paid-up.
 - The assumptions should be appropriate to the product being illustrated, have regard to the past experience, if available, and the Appointed Actuary's view of the expected future experience. Generally, these should be similar to the assumptions underlying the profit test carried out or the product filing with the regulator. However, if they are different, such differences should be justifiable.
 - The illustration should show two scenarios on investment return; a higher rate and a lower rate. The tables illustrating the benefit values should explicitly state the assumed rates of investment return. The current investment returns specified in APS-5 are 6% p.a. and 10% p.a. as specified by the Life

Insurance Council. However, these are 'maximum' only and the Appointed Actuary is free to use assumptions that are lower than these.

- For unit-linked plans, the illustrations should include full description of the various charges.
- Bonus rates in isolation can be misleading and therefore the Appointed Actuary may consider providing additional information on the rate of return in a suitable form taking care that the potential customers are not misled or misinformed.
- The assumptions underlying the illustrations should be reviewed at least annually.
- Options and benefits available in circumstances such as discontinuance of premiums before the policy acquires any value or after the policy acquires a paid up and/or a surrender value should be clearly spelt out together with the applicable conditions. [5]

iv) The policyholder IRRs for the money-back product is generally 40 to 50 bps lower than the erstwhile regular premium endowment product. In a with-profits policy, the maturity benefits are typically based on the projected asset shares. As such, a lower IRR for the money-back product would suggest that the projected maturity asset shares for this product is relatively lower as compared to that for the endowment product.

An attribution analysis can be performed to explain the difference in the projected asset shares for these two products by looking at various components of the asset shares. A consistent set of model points (covering demographic profile, premium amount etc.) should be used whilst carrying out this analysis.

The first step would be to start with the premiums accumulated at the illustrated gross investment returns for the two products. By definition (given the same premiums for the model points selected and the same assumed investment returns), this would imply the same returns for both products for this initial step. [1.5]

Subsequently, we could introduce various deductions to the asset shares one step at a time. Typically, these deductions would be in respect of:

- Commissions (both initial and renewal)
- Expenses (both initial and renewal)
- Cost of insurance (allowing for reinsurance cash-flows)
- Shareholder transfers
- Cost of capital
- Cost of guarantees
- Allowance for taxes

For each step, the incremental impact can be quantified by considering the changes to the projected asset share at maturity and the implied policyholder IRR associated with the introduction of the step. [2.5]

Likewise, if the Company's bonus philosophy allows for any additions to the asset shares in the form of miscellaneous profits, then these should be allowed sequentially, one step at a time. Such additions may include:

- Profits / losses on riders attached to the participating policies
- Profits / losses on any non-participating business written inside the participating fund, although this may not be possible in the current Indian regulatory environment
- Any lapse / surrender surpluses being written back to the asset shares of continuing policyholders

Again, for each step, the incremental impact can be quantified by considering the changes to the projected asset share at maturity and the implied policyholder IRR associated with the introduction of the step. [2]

Finally, a table such as the one below can be produced to summarize the results of the attribution analysis, clearly indicating the key drivers of the differential IRRs to the policyholders:

Particulars	Endowment Product	Money-back Product
Gross Returns	XX%	XX%
Initial Commission	(x%)	(x%)
Renewal Commission	(x%)	(x%)
Initial Expenses	(X%)	(X%)
Renewal Expenses	(X%)	(X%)
Cost of Insurance	(X%)	(X%)
Shareholder Transfers	(X%)	(X%)
Cost of Capital	(X%)	(X%)
Cost of Guarantees	(X%)	(X%)
Taxes	(X%)	(X%)
Misc. Profits	X%	X%
Lapse / Surrender Surpluses	X%	X%
Net policyholder IRR	XX%	XX%

The attribution analysis using asset shares as stated above can help in analyzing the reasons for the difference in policyholder IRRs for the two products. [2]

[7]

v) Possible reasons why the money-back product offers a lower IRR relative to the endowment product for comparable illustrations are set out below: [1.5 marks for each relevant point appropriately explained]

- Firstly, it is worthwhile checking the commission rates loaded in the premium rates for the two products. The maximum first year commission payable under the regulations is lower for limited premium payment plans (such as the money-back plan proposed to be introduced) as compared to regular premium payment plans for terms longer than 12 years (such as the 30 year regular premium endowment plan that the Company has been selling). However, if the actual commission loaded on and charged to the asset shares of the money-back product is higher than that for the endowment product, then this may lead to a lower policyholder IRR for the money-back product relative to the endowment product.
- Next, the initial expenses loaded in the premium rates for the endowment product should be reviewed vis-à-vis the money-back product. It is possible that the initial expenses loaded in the endowment product are lower relative to the money-back product, especially if the endowment product was priced long time ago and if the acquisition expenses / allocation basis has evolved over time. Alternatively, the strong volumes associated with the endowment product may have allowed the Company to charge a lower unit expense on this product relative to other newer products. Higher initial expenses loaded in the money-back product will ultimately pass through in the form of lower policyholder IRRs.

- The level of renewal expenses in the two products should also be compared. The money-back product is operationally more intensive relative to the endowment product – attributable to the processing of annual coupon payments and annual cash bonus payments after the end of the premium paying term. As such, this could lead to higher renewal expenses being reflected in the money-back product relative to the endowment product. If this is indeed the case, this can result in a lower policyholder IRR for the money-back product.
- Even if the commissions and expenses for the two products are similar, their effect (in percentage term) on policyholder IRRs can be quite different. In the endowment product the asset share will progressively increase over the policy term whereas in the money-back product the asset share will likely plateau or may even decline as the coupons and cash bonuses are paid out. As such, the commissions and expenses charged as a % of average asset share over the policy term may be significantly higher for the money-back product relative to the endowment product, leading to lower policyholder IRRs.
- If the death benefit significantly exceeds the asset shares, then this leads to a higher cost of insurance being charged to the surviving policyholders. For the endowment product, the asset share progression is steady and may lead to a situation wherein the asset share exceeds the guaranteed death benefit at some point during the policy term. Thereafter, the cost of insurance charged to the surviving policyholders may reduce. In the money-back product, the asset share will likely plateau after the premium payment term or may even decline as the coupons and cash bonuses are paid out. As such, the cost of insurance charged to the surviving policyholders may be higher than that under the endowment product, leading to lower policyholder IRRs.
- The actual mortality experience under the endowment product may also be different as compared to the expected mortality experience under the money-back product, especially if the products are distributed through different distribution channels and to customer segments with different profiles. If the mortality rates applied in the calculation of the cost of insurance charges are lower for endowment plan as compared to the money back plan, this may also lead to a higher cost of insurance charged to the asset shares of and lower policyholder IRRs for the money back plan.
- The level of guarantees under the two products should be reviewed. Typically, for a product that offers higher guarantees, the bonuses tend to be smaller. As such, the shareholder transfers and taxes charged to the asset shares may be smaller in comparison to products offering lower guarantees. On the other hand, the cost of guarantee charge applied to the asset shares may be higher for the money back plan, if the level of guarantees is higher than that for the endowment plan. The net impact of these may be higher for the money-back product as compared to the endowment product, leading to a lower policyholder IRR under the money-back plan.
- Review other items for comparativeness between the two products, as any differences may lead to different policyholder IRRs. Examples would include:
 - Treatment of miscellaneous profits e.g. riders
 - Treatment of lapse / surrender profits
 - Level of cost of capital charge reflected in asset shares
 - Differences in reinsurance arrangements leading to a differing impact on the asset shares for the two products.

[9]

vi) Various ways in which the experience can be reflected in the bonus declaration at the year-end and factors that the Company should take into account in deciding its approach are described in the following paragraphs.

If the Company has not been reflecting lapse profits in the asset shares of remaining policyholders and therefore the bonus rates declared, then any reduction in lapse profits should not impact the bonus rates

declarations anyway. Thus, whether or not the bonus rates should be impacted depend on the Company's philosophy in sharing of lapse profits with the remaining policyholders. [1]

1. Reduce reversionary bonuses

Reduced lapse surpluses can be reflected in the year-end bonus declaration by reducing the reversionary bonuses.

The Company has been seeing stable lapse surpluses in the with-profits segment every year. If these are being passed through to the asset shares of remaining policyholders and reflected in the reversionary bonuses declared in the past, then the Company may reduce the reversionary bonuses in order to pass through this experience to the policyholders. In this way, the declaration of bonus would be consistent with the historical practice.

On the other hand, the policyholders generally expect stable reversionary bonuses over time. As this particular matter appears to be a one-off market conduct incident of the agency channel of the Company, reflecting the impact of it by reducing the reversionary bonuses may be difficult to explain to the policyholders. Other things being equal, if the market conduct incident does not recur, the policyholders would see the reversionary bonus rates go back up a year later. From the perspective of meeting policyholders' reasonable expectations (PRE) to have smoothed bonus rates, such changes in reversionary bonus rates can be potentially confusing.

The Company will also have to take into account the immediate implications on marketing and new business, if it were to reduce the reversionary bonuses. A one off reduction in reversionary bonuses may adversely impact its new business volumes in the following year, especially if the competition does not have to lower their bonus rates. [4 marks for appropriately reasoned arguments]

2. Reduce terminal bonuses

Reduced lapse surpluses can be reflected in the year-end bonus declaration by reducing the terminal bonuses.

The Company has been seeing stable lapse surpluses in the with-profits segment every year. If these have been passed through to the policyholders in the form of reversionary bonuses, then the Company may continue to pass through the 'normal' level of lapse surpluses in reversionary bonuses declared, and reduce the terminal bonuses instead, to allow for the impact of this one-off market conduct incident. This may be acceptable from the PRE perspective.

On the other hand, reducing the terminal bonuses would imply that this will affect only those policyholders who are receiving their death or maturity claims in the following year. Depending on the extent of the 'lost' level of lapse profits, such reduction in terminal bonuses may have a disproportionate impact on the benefits payable to such cohorts of policyholders and therefore may not be equitable.

Another option would be to spread out the impact over several years to make it more equitable and fair.

Again, the Company will also have to take into account the immediate implications on marketing and new business if it were to reduce the terminal bonuses. [3 marks for appropriately reasoned arguments]

3. Charge to the with-profits estate, leaving bonuses unchanged

Instead of reducing the bonuses, the Company could simply let the impact of this one-off market conduct incident flow through to the with-profits estate, leaving the bonuses unchanged.

Given the one-off nature of this incident, charging this to the estate might be a good middle ground provided the with-profits fund is carrying a healthy level of estate. An argument can be made that one of the uses of the with-profits estate is indeed to absorb such unanticipated volatility and to provide cushion against the adverse impact on the policyholders' benefits.

The extent to which the estate can absorb this impact depends on the relative sizes of the loss and the estate itself.

Absorbing the loss would reduce the estate and would in turn affect the level of capital available to support future new business and to maintain investment freedom. [4 marks for appropriately reasoned arguments]

4. *Accept the impact in the shareholders' fund, protecting the with-profits fund*

From the standpoint of the with-profits policyholders, this would be the most beneficial outcome, albeit directly at the cost of shareholders.

An argument can be made that the market conduct incident is the culmination of control failures in distribution as well as product design and as such, the shareholders may wish to bear the loss instead of passing it on to the with-profits fund.

Pursuing this course of action may help avoid any adverse impact on the reputation of the Company that would arise if policyholders' bonuses or estate were to be adjusted.

On the other hand, this will set expectations going forward. Should such instances recur, the policyholders may expect shareholders to bear such losses in the future as well. This may not be viable all the time. In addition, such an action may set a precedent from a regulatory standpoint as well. As such, if this course of action is followed, the Company will have to clearly articulate this as a one-off gesture so as to set the right expectations.

Also, before considering this alternative, the Company may also need to assess its ability to declare intended level of bonuses based on the emerging statutory surpluses (which are lowered due to the lost lapse profits), given that based on the applicable regulatory circular, the Company may not be able to transfer funds into the with-profits funds for declaring bonuses after 12 years of commencement of its business.

[4 marks for appropriately reasoned arguments]

[16]

[50 Marks]

Solution 2:

[1 mark for each relevant decrement identified and impact on AoS explained]

i) The following decrements are possible from issuance until the ultimate cessation for unit-linked contracts in India:

1. Free-look cancellations

Policyholders may choose to cancel their policies within the free-look period, typically between 15-30 days after issuance. Those policies still within the free-look period and in-force as at the valuation date would typically have the following statutory reserves:

- As an in-force policy, statutory reserves equal to the unit reserve plus non-unit reserve (computed as present value of future non-unit cash-flows, subject to a minimum of zero), all subject to a minimum of surrender value.
- An additional free-look cancellation reserve, based on the Company's assessment of (prudent) probability of free-look cancellation.

Given that the decrement for free-look cancellation is not allowed for, the second of the above two reserve is not held. Hence, to the extent that there are actual free-look cancellations, a variance shall emerge within the analysis of surplus (AoS), equal to the amounts paid in excess of / lower than the in-force policy reserve.

2. 'In-force' to 'premium discontinuance' during the 'lock-in' period

Policyholders may discontinue premium payment during the first 5 years (i.e. the 'lock-in' period), at which time, their fund value, net of discontinuance charges shall be transferred to the 'discontinuance fund'. It is stated that only the mortality decrement is modelled, and hence, by implication, it appears that such discontinuance is not modelled for reserving purposes. To the extent that policies

discontinue premium payments in practice, there could be a significant variance within the AoS. This would arise given that the statutory reserves are held equal to unit plus non-unit reserves based on mortality decrement only, whereas the actual cash-flows during the year would be based on such premium discontinuance and transfer of surrender values into the discontinuance policy fund.

3. **Revival of 'premium discontinuance' policies:** Related to above, once premiums have been discontinued, the policy may be revived during the revival period. Allowance may be made for such policies within the valuation by way of a separate revival provision. In case the actual revivals are different to the assumed revivals within such a provision, there will be further contribution to the AoS.
4. **Death during 'lock-in' period after premium discontinuance:** Mortality decrements have been allowed for within the valuation assuming that all policies are effectively in-force (since no other decrements are modelled). However, a separate non-unit reserves may have been held for already discontinued policies allowing for mortality decrements. It is possible for the policyholder death to occur post premium discontinuance – in which case, death benefits shall be payable. To the extent that the allowance for mortality in the non-unit reserves calculation turns out to be different from the actual mortality experience, it may contribute to the AoS. This is likely to be a relatively less material source of variance in the AoS, but noted here for completeness.
5. **'Premium discontinuance' to 'surrender' (complete withdrawal without risk cover):** The policyholder may intimate intention to surrender during the 'lock-in' period, in which case the surrender value is transferred to discontinuance fund – as mentioned above – and the proceeds are payable at the end of the lock-in period. This is likely to be a material source of variance within the AoS given that surrender decrements are not allowed.
6. **'Premium paying' to 'premium discontinuance' after 'lock-in' period:** Similar considerations apply as stated earlier. However, this is applicable after the 'lock in' period and therefore considerations related to discontinuance fund do not apply. Treatment for such policies would depend on surrender terms and company practice with regards to policies that stop paying premiums but do not necessarily intimate surrenders.
7. **Revival of premium discontinued / surrendered policies (beyond 'lock-in' period) during permitted revival period:** As above, revival is possible within the revival period and AoS impact is likely to be driven by actual versus assumed experience within the revival provision.
8. **Surrender (complete withdrawal):** This is applicable for premium paying policies that request immediate surrender after the expiry of the 'lock-in' period. In general, reserves are floored to the surrender value, so any surrenders are likely to lead to a release of any excess reserves upon payment of surrender values and "surrender profits" within the AoS.
9. **Conversion to paid-up (full or reduced benefits depending on generation); death from paid-ups; partial or full withdrawal post paid-up (if permitted) and revivals from reduced paid-ups (if available)**
If conversion to reduced paid-up is permitted, upon premium discontinuance, further death or withdrawal is possible subsequently. Whilst death may be considered for policies already in paid up state, it is noted that future reduced paid-ups and the subsequent decrements of death / withdrawal / revivals may not be modelled currently – leading to another source of potential variance.
10. **In-force to death (regardless of lock-in period)**
This is listed as a decrement for completeness, noting that mortality is already modelled in the calculation of the non-unit reserves. Variance in AoS is possible due to actual mortality being different to the assumed mortality – as well as due to second order impact of not modelling other decrements, hence affecting the baseline exposed to risk between modelled and actual cash-flows.

11. **Surrender after completion of premium payment term (for limited pay only):** This is a special case of surrenders, a sub-set of overall surrender experience – for policies that surrender after payment of all due premiums in case of limited pay policies. It is likely that there are no surrender penalties for such cases.
12. **Partial withdrawals:** Unit-linked policies may allow partial withdrawals to meet certain financial needs, whilst continuing the policy. These could lead to variance in the AoS, since modelling assumes continuation of the full fund, which in reality may be depleted due to withdrawals and hence, affecting the overall level of charges reflected in the calculation of the non-unit reserves. **[10]**

ii) The following areas of expert judgment are likely to arise when determining the best estimate and valuation assumptions based on the information provided: [1 mark for each relevant point appropriately discussed]

1. **Number of years of experience to consider for setting assumption for initial policy durations:** It is noted that there is at least 10 years' worth of statistically credible data for policy durations up to 6 years. It is further noted that products that are currently available for new business are those launched post 2013. Moreover, although statistically credible data is available for a number of years, the experience itself has been volatile year-on-year. Typically, assumptions may be set by assigning greater weight to more recent experience as well as considering up to 3 – 5 years' worth of information. However, further care may be needed in the current case due to the observed year-on-year volatility. Therefore, judgement will be required with respect to the choice of number of observation years, in particular to consider whether to set assumptions based average of the (volatile) experience over a longer period of time or to set assumptions based on more recent observations (that may be more reflective of current and expected future experience).
2. **Relevance of experience:** Should experience in respect of policies sold before 2013 be considered? In case the target customer profile as well as distribution framework adopted to sell are not very different, a case may be made for using experience information over a longer observation period to estimate a "long term average" that smoothens the effect of volatility in individual years. Conversely, it may be argued that data from pre-2013 is redundant due to significant changes in product features, customer sophistication, target market, distribution framework etc. and that only more recent years' information should be used – particularly to set assumptions for early durations.
3. **Should past management actions and future plans and strategy be considered?** It is possible that some of the volatility (particularly in recent years) reflect management actions with respect to persistency management. For instance, it is possible that some (dis)incentives have been introduced for sales force related to (low)/high 13th month persistency that may have improved the 13th month persistency, but may have had some unintended consequences on observed experience in later years. Similarly, the management may have taken corrective actions, reacting to observed year-on-year trends, which may have caused the volatility in the observed rates. For example, a conscious effort by the management to influence experience may have played a role in the observed outcomes. Therefore, judgement would be needed with respect to how to allow for such past management actions when interpreting the volatility observed and whether, as an outcome of that, only the most recent experience (say, in past 1-2 years) should be considered relevant, as reflecting the current management philosophy for persistency management. Similarly, appropriate adjustments may be required to best estimate assumptions to reflect the current expectations, in case current management actions / future plans / strategies are different to those adopted historically.
4. **Reliance on internal versus external data points:** For both early and later policy durations, the observed internal experience appears to show some deficiencies: for shorter policy durations, there is credible

experience but it is volatile; for longer durations, there is limited experience, yet suggesting stable rates. In view of this, judgement may be required regarding the extent to which internal data should be relied upon and whether any external sources of data / information should be considered to set appropriate assumptions. These external sources of information may be available from industry studies or research, if any or from consultants who may be able to provide broader industry perspectives.

5. **Consideration for longer duration assumptions for post-2013 products, where experience is not available:** The information provided indicates that for post-2013 products, the actual experience in early policy durations has been worse than anticipated at pricing stage. Therefore, this could indicate that initial pricing assumptions were on the lower side and overall, all assumptions could be revised upwards reflecting overall worsening of actual versus expected ratios – including assumptions for later durations. Alternatively, another perspective may be to maintain longer duration assumptions unchanged from pricing / prior assumptions due to non-availability of any additional information / data-points for these durations. A further revised view on expected longer term experience may be needed, given the shorter duration experience to date – i.e. whether all “poor quality risks / business” has already exited and therefore, the residual policies remaining are likely to be more persistent than previously assumed.
6. **What should be the appropriate level of granularity at which assumptions are set:** Not much information is provided in respect of any possible variations in actual as well as expected experience for factors other than policy duration. For example, the experience may vary by products, by different distribution channels, by homogenous categories / cohorts (i.e. say, by age / gender / premium payment mode / case size or other relevant factors). It is possible that, when experience is viewed at a more granular level, this historic volatility observed in the overall experience is lower. It is also possible that the observed volatility potentially represents changes in business mix across such cohorts over time, whereas the underlying experience within the relevant cohort may well be comparatively stable. Either way, the level of granularity for both undertaking the experience investigation as well as setting assumption could be an important area of judgement.
7. **Unit of measurement of experience:** Certain degree of judgement may also be needed to determine whether to anchor assumptions to experience weighted by policy count or that weighted by premium, particularly in case the underlying variation in experience between the two is significant. Whilst policy-weighted experience might be more relevant for modelling policy-level decrements, premium-weighted experience may be preferred in case trying to reflect more material aspects of the business from an amounts perspective.
8. **Level of margins for prudence:** When setting the valuation assumptions, of utmost importance might be the judgement related to the level of prudent margins to be incorporated. Whilst minimum margins as specified in APS 7 would determine the floor, further arguments may be made for making an allowance for higher than the minimum prescribed margin for adverse deviations, in light of the observed volatility (for early durations) and lack of experience (for later durations). Additionally, judgement may be needed to assess deviation from experience when setting the best estimate assumptions themselves and whether any implicit or explicit margins are required therein. Finally, a related consideration might be to assess whether uni-directional margins should be set or whether the direction of any margins (i.e. higher or lower relative to best estimate assumption) should be set separately for each duration depending on the statistical credibility of the underlying experience and the likely impact on the results.
9. **Overall judgements related to materiality and proportionality:** Each of the areas discussed above may need to be tempered with overall judgments related to materiality and proportionality of the individual choices made. It is possible that in the absence of any material guarantees, the non-unit reserves may be

a low proportion of the overall reserves under the unit-linked business. Given this, a pragmatic approach may be preferred in setting the best estimate and valuation assumptions. [9]

iii) [0.5 mark for each relevant point made, 1.5 marks for each relevant point made and appropriately discussed]

The most significant consideration for the Company in assessing the proposed product is the fact that it may have limited experience of pricing and selling of non-participating ('non-par') protection products, given its historic unit-linked focus. Therefore, whilst making its initial foray into developing a new line of business, the Company may consider a product design that limits its risks while providing opportunity for scalability based on any emerging learnings. [0.5]

Overall, the factors that need to be considered when determining whether or not the proposed product design is suitable, are discussed below:

1. **Product structure and profitability:** This is likely to be one of the primary criteria for internal product approvals. In the current operating environment, whilst it is possible that non-par protection products may appear to be more profitable than unit-linked products, the Company may need to closely assess the proposed design to ensure that it fully understands sources of such profits. That is - whether the profitability of the contract is driven by risk margins between priced mortality rate reflected in the premiums charged and the expected mortality (i.e. in case there is an explicit margin within the premium rates for mortality risk) or whether it is due to other factors such as lapse profits / reinsurance profits / investment spread or potentially expense spreads. Further, the product design itself may need to be looked at – both with respect to the timing and the amounts of the benefit structure – to ensure that the profitability is optimized. For instance, the proposed limited pay structure may need to be studied for longer or shorter premium payment terms to assess impact on profitability. Similarly, the benefits offered may need to be calibrated to ensure appropriate balance of various sources of profits as set out above.
2. **Marketability:** The insurer needs to carefully consider the potential demand for the product features – both in the current operating environment as well as likely future developments. For instance, is a whole-of-life product design likely to be more attractive / marketable than a fixed term protection with or without return of premium options? Similarly, should the Company introduce certain options within the product design to make it more widely marketable? For instance, to make the product more marketable, consider options with respect to premium payment (i.e. single premium / limited pay or regular pay rather than fixing it as a 'limited pay only' product) as well as in respect of benefit terms (e.g. fixed term or whole of life, options with respect to how death benefits are paid out – i.e. in lump sum or instalments) etc.
3. **Suitability to meet customer needs for defined target market:** The Company's existing customer base may largely consist of customers who have purchased a unit-linked product. Whilst offering a protection product does offer the ability to the Company to meet different customer needs at different life stages, it needs to consider carefully whether the product design and features indeed meet the needs of its customers or whether there is a risk of mis-selling to its target market – where certain customers may buy into the product without fully understanding its merits or applicability to their own life-goals and priorities.
4. **Competitiveness:** Given that the Company has not historically sold much protection business, it is possible that the product design and pricing builds in some buffers to compensate for the lack of prior experience. However, care is needed to ensure that this does not make the product uncompetitive. At the same time, the Company may consider its ability to charge slightly higher level of premiums to its 'captive' customer base derived through superior brand value / goodwill, ease of onboarding or any other such factors and price its product accordingly, reflecting its relative competitive strengths.

5. **Distribution method and remuneration:** The information provided indicates that the Company's existing distribution channels, tied agency, would continue to be used to sell the proposed new protection product. The agency force is currently accustomed to selling unit-linked products. Therefore, care is needed not to make the product design overly complex so as to enable effective training of the agents on a completely new product. Moreover, the company would also need to consider the commission structure and related incentives – and how these compare with the existing unit-linked products, to ensure that any disparity does not create unintended (dis)incentives among the agents.
6. **Financing / capital requirements:** The product design would need to be capital efficient. A limited pay, whole of life protection product is likely to have “front-loaded” income structure. Therefore, it is possible to design the premiums/benefits in a manner that result in low or minimal reserving / capital requirements initially. Ideally, the product design may be tweaked to ensure that any new business strains / capital requirements are reduced as much as possible.
7. **Onerousness of options and guarantees:** Based on the information provided, it is not clear whether there are any material options or guarantees provided in the proposed product design. Although the presence of any such options / guarantees is likely to make the product more attractive from a policyholder perspective (and therefore could be considered for marketing / competitive reasons), at the same time, financial cost of these should be considered from Company's perspective to ensure that these are not too onerous. Extreme stress testing and / or stochastic analysis may be undertaken to judge onerousness.
8. **Sensitivity of profits:** As mentioned earlier, it is important to understand the sources of profits within the product design. At the same time, the Company should consider how sensitive the profits are to each material variable including both economic and operating outcomes. This may be particularly relevant in the current case given long term nature of the contract (being whole of life) and the fact that the Company is new to this line of business. Therefore, the Company should minimise its risks by having a product design where profits are not overly sensitivity to some particular variables.
9. **Extent of cross subsidies:** The pricing of the product is likely to introduce some degree of cross-subsidies, e.g. between larger / smaller case sizes, or between ages, smoking status etc. This may be necessary for either smoothing or commercial consideration in certain cases. However, the degree of cross-subsidies should be carefully considered against the risk of new business mix deviating from the optimal mix and the ability of the Company to finance adverse variation in case of business mix changing.
10. **Administration systems:** This may be an important consideration given a completely new line of business. Does the Company have administration systems that are easy to modify to on-board a new line of business? In particular, given the whole of life nature of contracts, are any special adjustments needed to the systems that may be overly cost-prohibitive? Furthermore, care would be needed so as not to make the product design too complex such that the system changes needed to implement may become unnecessarily tedious and costly.
11. **Service standards:** It is possible that a number of specialist operational and underwriting staff may be needed, particularly to manage claims / frauds on protection portfolio. Therefore, in developing this product, the Company should carefully consider the quality of service it is able to provide and to ensure that there is no dilution of service standards for this new line of business.
12. **Company reputation:** The Company has previously built a reputation as a unit-linked provider. Therefore, does the new product offering fit in with its reputation or could likely cannibalise its current brand recall? Prior marketing, sales and branding materials may need to be revisited to ensure that the new product does not create confusion in the minds of customers, distributors as well as the broader stakeholders as to the Company's focus areas and strategy.

13. **Treating customers fairly:** The product design may need to be carefully reviewed from customer perspective. Particularly, the terms related to surrender values or reduced paid up values in case of premium discontinuance should be set to ensure that the benefits offered are fair and reasonable.
14. **Level of risk:** The product designing and pricing should be carried out in a manner that introduces an acceptable level of risk for the Company. Given the long term protection nature of the contracts, the Company should review whether the mortality rates priced into premiums are too aggressive in light of any expected changes in demographic experience over the long term. The Company should also ensure that there is no over-reliance on lapse profits or reinsurance profits.
15. **Underwriting philosophy:** This may be an important focus area given the Company's foray into protection product from unit-linked. The product design needs to be consistent with underwriting philosophy particularly, in case any simplified or straight-through underwriting is intended for all or some proportion of business. Further care needs to be taken to ensure that possible fraud is checked at the underwriting stage itself, as the Company may not be accustomed to the level of underwriting control required under protection business given historic unit-linked focus.
16. **Reinsurance terms and other benefits:** As the Company does not have much experience of writing protection business, it may need to rely on reinsurance support in a number of key areas including underwriting, contract wordings, initial rate settings etc. Moreover, given the competitive environment, the reinsurance terms may be favourable and end up as a "source of profit" as opposed to a cost within the overall pricing basis. This may need to be carefully considered for sustainability. Additionally, the degree of alignment between reinsurance rates and product design / pricing assumptions may also need to be considered. For example, if the reinsurance rates vary by gender / smoker status, then the Company may consider the pricing assumptions that vary by similar factors too.
17. **Taxation:** In the current tax scenario, it is possible that the Company is enjoying certain favorable tax breaks / deductions within its unit linked portfolio particularly, in case of investments equities from the unit-linked funds that provide tax free dividends. When launching the new product, careful consideration may be necessary to study how it impacts the overall tax position of the Company. Additionally, the product design should be such that the policyholders are able to avail any tax incentives available.
18. **Legal and regulatory constraints:** There are a number of regulatory requirements pertaining to product design and minimum benefits as set out within the IRDAI (Non-linked insurance products) Regulations, 2013. These must be considered in the development and pricing of the product. [18]

iv) [1 mark for each of the above main points identified]

As set out in the IRDAI (Reinsurance) Regulations, 2018, the following requirements need to be considered when setting out the reinsurance arrangements for the proposed new product:

1. **Minimum retention limit:** This is specified to be 25% of sum at risk, given the pure protection nature of the product. However, the objective in setting out the reinsurance program should be to retain maximum possible cover based on the Company's financial strength that protects the interest of the policyholders at a reasonable cost and helps in simplifying the administration;
2. **Reinsurance program:** This is required to be Board approved and submitted to the IRDAI annually. The Company should be able to justify its reinsurance strategy taking into account the business mix and risks undertaken;
3. **Catastrophic risks:** The Company should assess its catastrophic risk exposure against the new product and consider whether separate catastrophe reinsurance cover is necessary;

4. Type of reinsurer and limits: The Company should have a preference for a domestic reinsurer and should consider placing business with cross-border reinsurers (CBR) provided the criteria set out for CBR are met. Also, the Company is required to seek best reinsurance terms from all Indian reinsurers and at least 4 foreign reinsurance branches, all with credit rating of at least A- or higher. There are also maximum cession limits specified for each CBR, if used.

5. Information submission: The Company would be required to submit information pertaining to its reinsurance transactions in a year to the IRDAI. **[5]**

v) Based on the information provided, it is evident that the reinsurer rates imply a lower expected mortality cost than the current mortality experience of the Company.

The cost to the Company within the proposed reinsurance arrangement is the required payment of reinsurance premiums at the rate of 50% of IALM 2006-08 for the portion of business that it cedes. The benefit would be any reinsurance recoveries made for the amount of risk ceded based on actual mortality experience of the Company. In addition, the Company may enjoy certain softer benefits from the reinsurance arrangement such as expertise needed to underwrite / price for / deploy or manage the new protection product, as well as perhaps any systems support that the reinsurer may provide.

In case the Company treats the financial cost and benefit based on its own experience at face-value, then within the pricing of the new product, it may effectively assume 100% of IALM 2006-08 as its 'best estimate' mortality rates for reinsurance recoveries, while reinsurance premiums are payable at 50% of IALM 2006-08. This could lead to a significant "reinsurance profit" within the pricing due to the recoveries being higher than reinsurance premium payments.

The Company needs to carefully consider whether this implied reinsurance benefit is appropriate, particularly given its own lack of experience in selling protection business. Additionally, it may need to consider whether the reinsurance is applicable only to a sub-section of its portfolio (e.g. only in respect of higher case size policies that exceed the specified retention limit). In case there is material variation in the expected mortality experience for the proportion of business that is likely to be reinsured and that which is likely to be fully retained, then this may need to be allowed for in setting the 'best estimate' mortality assumption by setting assumptions that are dependent on the size of the policy.

Additionally, it is possible that the reinsurer may be pricing based on its overall industry experience and that the Company's own experience for its target customer base / agency distribution channel is genuinely poorer. In this case, the reinsurer may run the risk of having "mis-priced" relative to the Company's target customer segment / distribution channel. If the Company is convinced about this, some benefit of the low reinsurance rates may be reflected in pricing. However, the Company may need to exercise care in reflecting such benefit as the reinsurance arrangement may not be sustainable and the reinsurer may revise its pricing and/or withdraw the arrangements in the future if its own emerging experience leads to potential losses.

Furthermore, in case it does turn out that the ongoing recoveries from the reinsurer are materially higher than the priced rates, there is a risk to the Company of significant disputes with the reinsurer. For example, the reinsurer may attempt to minimise losses by scrutinizing each claims and potentially repudiating claims, which the Company in turn must honor due to regulatory or other considerations. Also, although very unlikely given the high credit rating of the reinsurers, in extreme cases, such sustained level of high losses may run the risk of default by the reinsurer. The Company may need to consider such risks and allow for the same in pricing the product, by at least testing the likelihood and financial implications of such a scenario.

A further consideration that may be required is how much of the benefit arising from favorable reinsurance rates should be “passed back” to the customers by way of lower premiums and how much of it “retained” within the Company. For example, when setting the premium rates, the Company may choose to use its own experience at 100% mortality rates. However, when undertaking profit testing, the “best estimate” mortality assumption may be based on the reinsurance rates, thereby achieving a high level of profit margin for the Company. Alternatively, in case the Company uses reinsurance rates for determining office premiums, the benefit of lower reinsurance rates would be passed on to the policyholders via lower premium rates.

Another significant benefit of the reinsurance arrangement would be the ability of the Company to lower its reserves and capital requirements by utilizing the maximum credit permissible under the IRDAI regulations. This may help lower the cost of capital reflected in pricing of the product. Thus, the cost / benefit of reinsurance arrangements need to be considered by taking into account both cash benefits (i.e. the difference between reinsurance premiums and recoveries) as well as the benefit of lowering of reserves and capital requirements.

[1.5 marks for each point commented upon]

[8]

[50 Marks]
